

Neural Networks for Data Science

Master's Degree in Data Science

Lecture 4: Fully-connected neural networks

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Feedforward neural networks

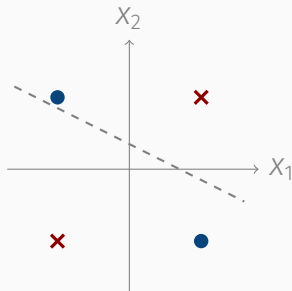
Limitations of linear models

For binary classification, a linear model separates the two classes using:

$$\mathbf{w}^T \mathbf{x} + b = 0.$$

Such a **linear boundary** can only separate relatively simple datasets.

Can we **learn a representation** of the data where a linear model is sufficient?



A simple non-linearly separable dataset (XOR).